

Software Requirements Specification (SRS)

Web-Based Vehicle Auction Platform

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements of the **Web-Based Vehicle Auction Platform** intended for a vehicle exporter operating from Japan. The SRS serves as a contractual and technical reference between stakeholders (client, supervisor, users and developer) and will be used as the baseline for system design, implementation, testing, and evaluation.

1.2 Intended Audience

- Client (Tapro Japan Co. Ltd.)
- Project Supervisor

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
Auction	Time-bound bidding event for a vehicle
Bid	Monetary offer placed by a buyer
Verified Buyer	Registered user approved by admin
Admin	System administrator with full privileges
SRS	Software Requirements Specification
FR	Functional Requirement
NFR	Non-Functional Requirement

2. Overall Description

2.1 Product Perspective

The system is a **standalone web-based application** designed to replace manual vehicle auction coordination conducted via messaging platforms.

It follows a **three-tier architecture** consisting of:

- Frontend (User Interface)
- Backend (Business Logic and APIs)
- Database (Persistent Storage)

2.2 Product Functions (High-Level)

- User registration and authentication
- Buyer verification by admin
- Vehicle listing and browsing
- Auction scheduling and lifecycle management
- Bidding and highest-bid tracking
- Administrative monitoring and reporting

2.3 User Classes and Characteristics

User Class	Description
Guest User	Can browse vehicles but cannot bid
Registered User	Can view details but cannot bid until verified
Verified Buyer	Can place bids on active auctions
Admin	Manage users, vehicles, auctions, and bids

2.4 Operating Environment

- Web-based system accessed via modern browsers
- Backend server running Node.js
- PostgreSQL relational database
- Deployed on a standard web hosting or cloud environment

2.5 Design Constraints

- No payment gateway integration
- No shipping or logistics handling
- Compliance with secure coding practices

2.6 Assumptions and Dependencies

- Users have stable internet access
- Admin manually verifies buyers
- System operates within moderate user load

3. System Features

3.1 User Management

- Registration using email and password
- Secure login/logout
- Role-based access control
- Admin verification workflow

3.2 Vehicle Management

- Admin-managed vehicle listings
- Image and specification display
- Search and filtering functionality

3.3 Auction Management

- Creation of auctions with start/end times
- Automatic auction state transitions
- Visibility of auction status

3.4 Bidding System

- Bid placement by verified users only
- Highest bid detection
- Bid history tracking
- Prevention of late bidding

3.5 Administrative Dashboard

- User verification controls
- Auction monitoring
- Bid review and winner selection
- Basic reporting

4. Functional Requirements

(Each requirement is **atomic, testable, and traceable**)

4.1 Authentication & User Management

- **FR-01:** The system shall allow users to register using email and password.
- **FR-02:** The system shall authenticate users using secure credentials.
- **FR-03:** The system shall allow only users whose accounts have been verified by an administrator to place bids on active auctions.
- **FR-04:** The system shall allow admins to approve or reject user accounts.

4.2 Vehicle Management

- **FR-05:** The system shall allow admins to add, update, and delete vehicle listings.
- **FR-06:** The system shall display vehicle details including images and specifications.

4.3 Auction Management

- **FR-07:** The system shall allow admins to create auctions with defined start and end times.
- **FR-08:** The system shall automatically update auction status based on time.
- **FR-09:** The system shall prevent bids on closed auctions.

4.4 Bidding

- **FR-10:** The system shall allow verified buyers to place bids on active auctions.
- **FR-11:** The system shall validate bids against minimum increment rules.
- **FR-12:** The system shall record all bids with timestamps.
- **FR-13:** The system shall identify and display the current highest bid.

5. Non-Functional Requirements

5.1 Security

- **NFR-01:** Passwords shall be stored using cryptographic hashing.
- **NFR-02:** The system shall prevent unauthorized access via role-based controls.

5.2 Performance

- **NFR-03:** The system shall process bid submissions within acceptable response time under normal load.

5.3 Reliability & Availability

- **NFR-04:** The system shall maintain data consistency for all bidding transactions.

5.4 Usability

- **NFR-05:** The system shall provide a user-friendly web interface accessible on desktop and mobile browsers.

5.5 Scalability

- **NFR-06:** The system shall support future expansion without major architectural changes.

6. External Interface Requirements

6.1 User Interface

- Web-based graphical user interface
- Separate interfaces for admin and buyers

6.2 Software Interfaces

- RESTful APIs between frontend and backend

6.3 Database Interfaces

- Relational database storing users, vehicles, auctions, and bids

7. Out-of-Scope Features

- Online payments
- Shipping and logistics management
- Advanced analytics and AI-based recommendations

8. Future Enhancements

- Payment gateway integration
- Automated notifications via email/SMS
- Advanced auction analytics